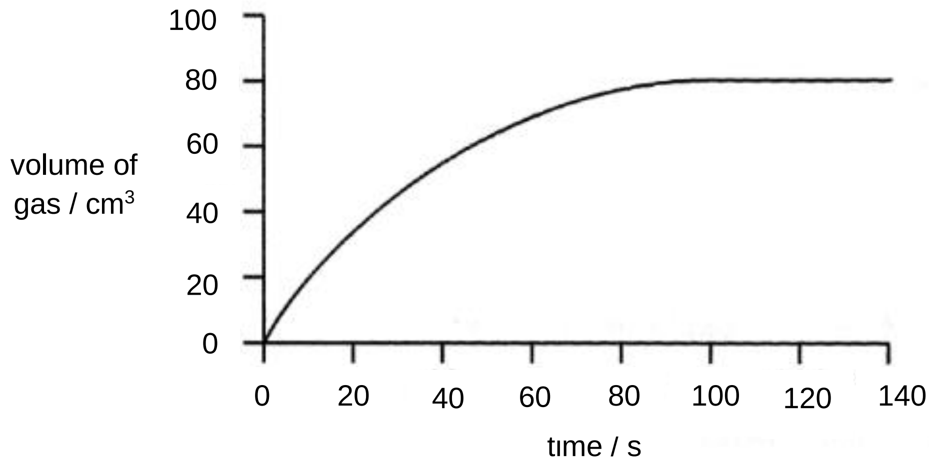
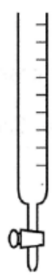
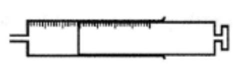
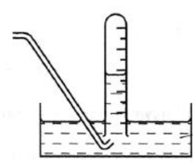
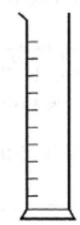
1 Zinc reacts with excess dilute hydrochloric acid.
The graph shows the volume of hydrogen gas evolved at 20 second interval until the reaction had finished.

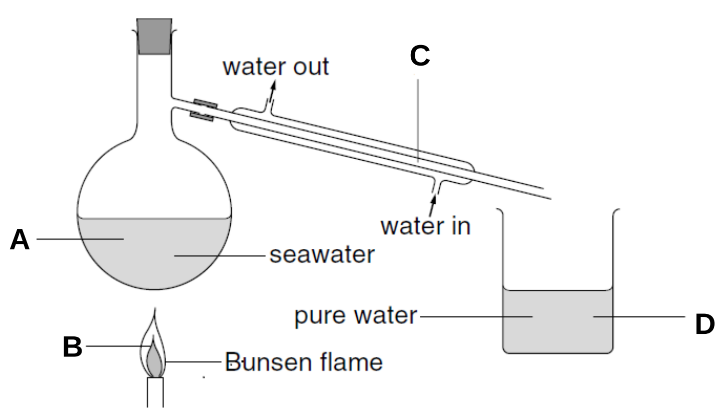


Which piece of apparatus would be suitable for measuring the volume of hydrogen gas evolved?

A	B	C	D
 50 cm^3 burette	 100 cm^3 gas syringe	 50 cm^3 graduated tube	 100 cm^3 measuring cylinder

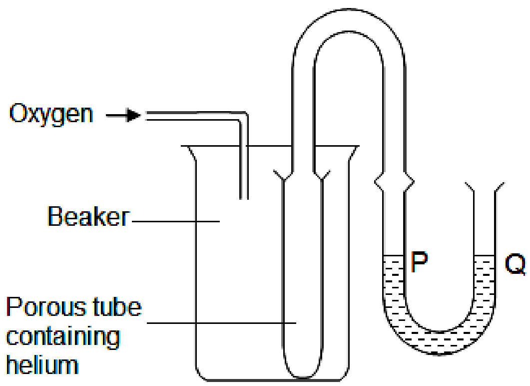
2 The diagram shows how to obtain pure water from seawater.

Which point do water molecules lose energy?

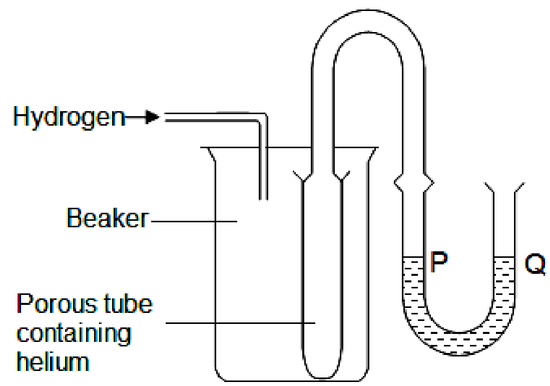


3 Experiments 1 and 2 are set up to demonstrate the diffusion of gases.

What would happen to the water levels at P and Q in both experiments?



Experiment 1



Experiment 2

	Experiment 1	Experiment 2
A	P is higher than Q	P and Q remain the same
B	P is higher than Q	Q is higher than P
C	P and Q remain the same	P and Q remain the same
D	P and Q remain the same	P is higher than Q

4 The reaction between aqueous cobalt(II) nitrate and aqueous potassium carbonate is shown.

$$\underset{\text{red}}{Co(NO_3)_2\ (aq)} + \underset{\text{colourless}}{2K_2CO_3\ (aq)} \rightarrow \underset{\text{purple}}{CoCO_3\ (s)} + \underset{\text{colourless}}{2KNO_3\ (aq)}$$

Which method could be used to separate the products?

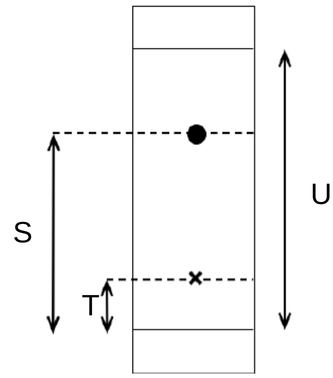
A chromatography

B crystallisation

C distillation

D filtration

5 A substance is placed on a piece of filter paper at the point marked X and placed in a solvent for some time. The resulting chromatogram is obtained.



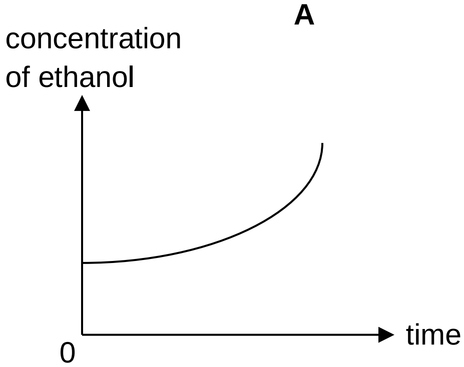
What is the R_f value of this substance?

A $\frac{S}{U}$ **B** $\frac{T}{U}$ **C** $\frac{S-T}{U}$ **D** $\frac{S-T}{U-T}$

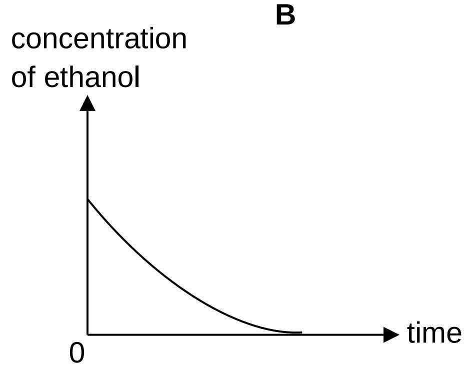
6 Fractional distillation is used to obtain pure ethanol (boiling point 78 °C) from a dilute ethanol solution.

Which of the following graphs best shows the change in concentration of ethanol in the boiling flask as distillation proceeds?

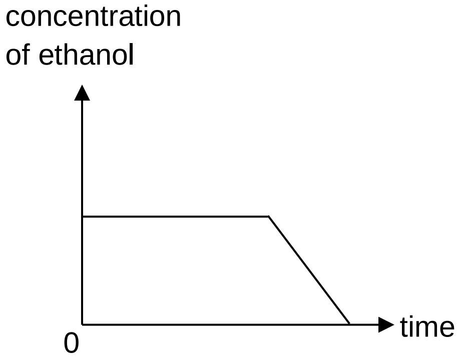
A



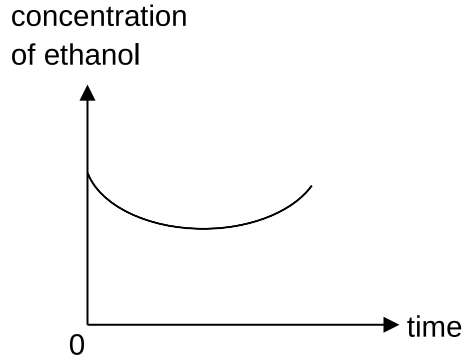
B



C



D



7 The table gives some information about particles S and T.

particle	number of protons	number of neutrons	electronic configuration
S	7	7	2,8
T	16	16	2,8,8

What are particles S and T?

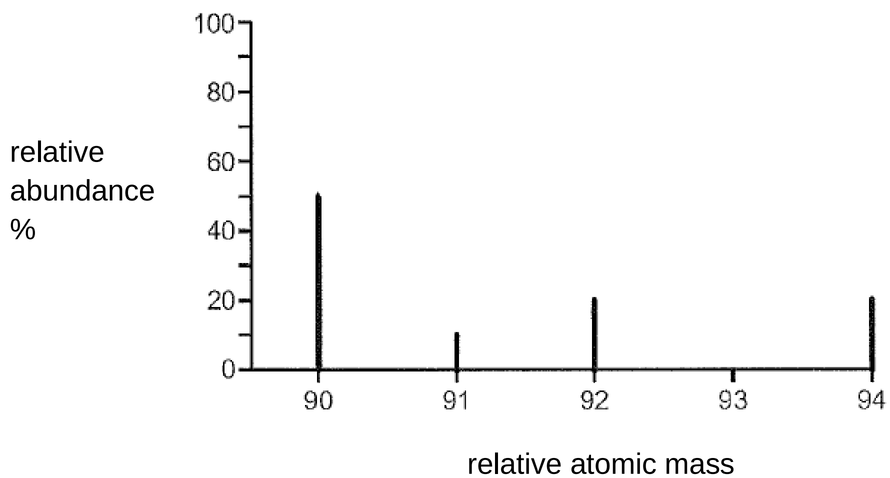
A atoms of metals

B atoms of noble gases

C ions of metals

D ions of non-metals

8 An element Z consists of four isotopes.
The graph shows the relative abundance of the isotopes of Z.



What is the relative atomic mass of element Z?

A 91.0

B 91.3

C 91.8

D 92.0

9 Which of the following statements about the particles $^{23}Na^{+}$ and ^{20}Ne is correct?

A Both Na^{+} and Ne have 10 electrons.

B Na^{+} has 3 more neutrons than Ne.

C Ne has more electrons than Na^{+}.

D Na^{+} has 2 more protons than Ne.

10 Covalent bonds are formed when atoms share electrons.

How many electrons are provided by atom of each element when they form covalent compounds?

	oxygen	fluorine	carbon
A	1	2	3
B	1	2	4
C	2	1	3
D	2	1	4

11 Element V has n protons and element W has (n+2) protons.
Element V forms ions with charge of 3–.

Which statement correctly describes the compound formed between V and W?

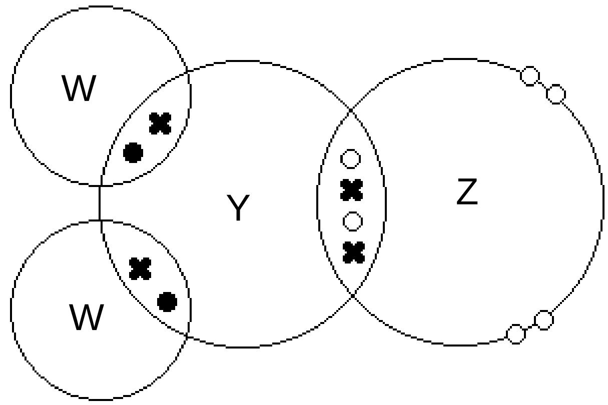
A It is a covalent compound with the formula VW_3.

B It is a covalent compound with the formula V_2W_3.

C It is an ionic compound with the formula VW_3.

D It is an ionic compound with the formula V_2W_3.

12 The 'dot-and-cross' diagram of a compound is as shown.
Only valence electrons are shown.



What are the elements W, Y, and Z?

	W	Y	Z
A	hydrogen	boron	nitrogen
B	hydrogen	carbon	oxygen
C	sodium	boron	nitrogen
D	sodium	carbon	oxygen

13 The table shows the physical properties of four unknown substances L to O.

	melting point /°C	boiling point /°C	electrical conductivity in		solubility in water
			solid state	liquid state	
L	163	440	poor	poor	insoluble
M	580	1800	poor	good	soluble
N	1828	2380	poor	poor	insoluble
O	1553	2980	good	good	insoluble

Which of the following statements about the four substances is correct?

A Substance L has a simple molecular structure with weak covalent bonds between molecules.

B Substance M has a giant lattice structure with mobile electrons held by strong electrostatic forces.

C Substance N has a giant covalent structure with strong intermolecular forces between molecules.

D Substance O has a giant metallic structure with mobile electrons.

14 The relative molecular mass, M_r, of copper(II) sulfate, $CuSO_4$, is 160.

What is the percentage by mass of water in copper(II) sulfate crystals, $CuSO_4.5H_2O$?

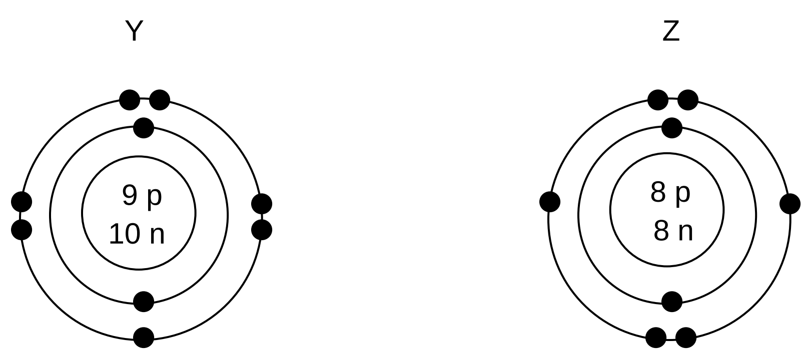
A $\dfrac{18 \times 100}{160}$

B $\dfrac{5 \times 18 \times 100}{160}$

C $\dfrac{18 \times 100}{160 + 18}$

D $\dfrac{5 \times 18 \times 100}{160 + (5 \times 18)}$

15 The diagram shows the electronic structures of the atoms of two elements, Y and Z.



When these two elements combine to form a compound, what is the mass of 1 mol of the compound?

A 26

B 35

C 51

D 54

16 The formation of metallic oxides involves the transfer of electrons from a metal atom to oxygen atoms.

In the formation of one mole of which metallic oxide do the metal atoms **not** transfer exactly two moles of electrons?

A aluminium oxide

B iron(II) oxide

C magnesium oxide

D sodium oxide

17 20 cm^3 of an aqueous 1.0 mol / dm^3 solution of the hydroxide of metal X, exactly neutralises 40 cm^3 of aqueous 0.25 mol / dm^3 sulfuric acid.

What is the formula for the sulfate of X?

A X_2SO_4 **B** XSO_4 **C** $X_2(SO_4)_3$ **D** $X(SO_4)_2$

18 The chemical equation for the complete combustion of methane is as shown.

$$CH_4\,(g) + 2O_2\,(g) \rightarrow CO_2\,(g) + 2H_2O\,(g)$$

50 cm^3 of methane gas is mixed with 40 cm^3 of oxygen gas in a sealed vessel and burnt.

What is the volumes of the gases remaining, at the original temperature and pressure?

	volume of methane / cm^3	volume of oxygen / cm^3	volume of carbon dioxide / cm^3	volume of steam / cm^3
A	0	0	50	40
B	0	20	20	40
C	30	0	20	40
D	40	40	40	40

19 When 0.5 g of impure limestone (calcium carbonate) was added to excess hydrochloric acid, 90 cm^3 of carbon dioxide gas was evolved.

What is the percentage purity of the limestone?

A 15%

B 50%

C 75%

D 85%

20 Which liquid is the best conductor of electricity?

A aqueous ethanoic acid with a concentration of 1 mol / dm^3

B aqueous hydrogen chloride with a concentration of 1 mol / dm^3

C pure ethanoic acid

D pure liquid hydrogen chloride

21 Which of the following must be true for all acids?

1 Their pH value is less than 7.

2 They are completely ionised when dissolved in water.

3 They react with all metals to give hydrogen gas.

4 Dibasic acids are stronger than monobasic acids.

A 1 only

B 1 and 2 only

C 1, 2 and 4 only

D 1, 2, 3 and 4

22 The oxides of three elements, T, U and V, are added to water.

	oxide of T	oxide of U	oxide of V
water added	dissolved to form a solution of pH 2	insoluble	dissolved to form a solution of pH 13

The oxide of U is white in colour.

What are T, U and V?

	T	U	V
A	calcium	aluminium	sulfur
B	calcium	copper	sulfur
C	sulfur	aluminium	calcium
D	sulfur	copper	calcium

23 The diagram shows the colours of the indicators, methyl orange and methyl red, at different pH values.

<table>
<tr><th>pH</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th></tr>
<tr><td>colour of methyl orange</td><td colspan="2">red</td><td colspan="3">yellow</td></tr>
<tr><td>colour of methyl red</td><td colspan="4">red</td><td>yellow</td></tr>
</table>

The pH of different solutions is given in the table.

solution	W	X	Y	Z
pH	2	3	5	6

In which solution/s will both indicators be yellow?

A W and X

B X and Y

C Y and Z

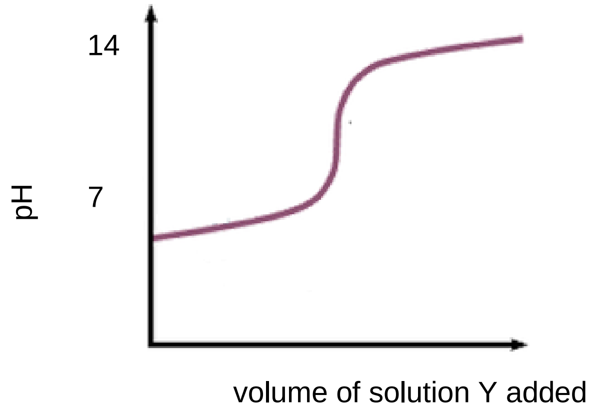
D Z only

24 How many different soluble chlorides in total could be prepared by the reaction of dilute hydrochloric acid with the following substances?

- copper
- magnesium
- silver nitrate
- zinc carbonate

A 1 **B** 2 **C** 3 **D** 4

25 Solution Y was added to aqueous solution Z.
The changes in pH of the mixture are shown in the graph.



What could Y and Z be?

	Y	Z
A	aqueous ammonia	hydrochloric acid
B	sodium hydroxide	hydrochloric acid
C	aqueous ammonia	ethanoic acid
D	sodium hydroxide	ethanoic acid

26 Five students were preparing potassium chloride by titration.
They each pipetted 25.0 cm^3 of potassium hydroxide solution into a conical flask and titrated with dilute hydrochloric acid, using methyl orange.
The results are shown in the table.

student number	1	2	3	4	5
volume of hydrochloric acid used / cm^3	20.30	20.10	20.60	20.40	22.80

Which statement could explain the result obtained by student number 5?

A The burette was washed with hydrochloric acid.

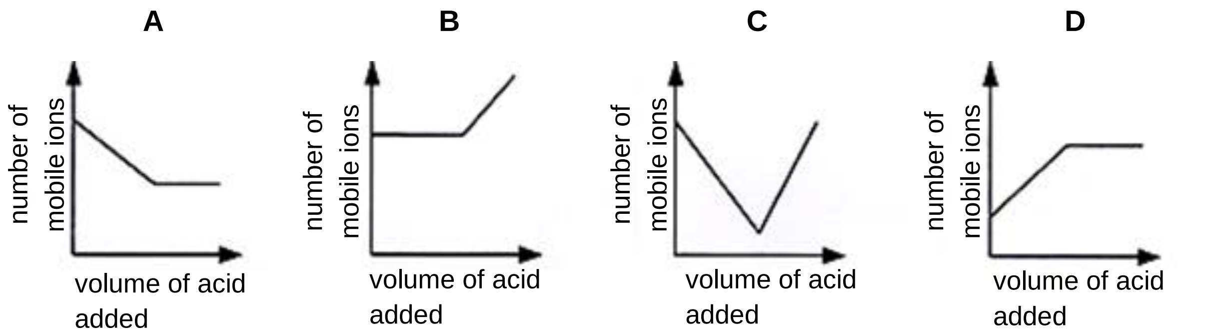
B The pipette was washed with potassium hydroxide solution.

C The conical flask was washed with deionised water.

D The conical flask was washed with potassium hydroxide solution.

27 Dilute sulfuric acid was added to aqueous barium hydroxide until the acid was in excess.

Which graph best represents the variation in the total number of mobile ions in solution?



28 Which option shows the appropriate reagents used for preparing the named salt?

	salt	reagents
A	barium chloride	barium sulfate + sodium chloride
B	lead(II) sulfate	lead(II) chloride + ammonium sulfate
C	magnesium chloride	magnesium + hydrochloric acid
D	sodium chloride	sodium metal + hydrochloric acid

29 Magnesium sulfate is made by reacting magnesium carbonate with dilute sulfuric acid.

The method used is shown.

step 1: Add excess magnesium carbonate to dilute sulfuric acid.

step 2: Filter the mixture and collect the filtrate.

step 3: Heat the filtrate in an evaporating dish until crystals start to form.

step 4: Leave the solution formed to cool.

Which substances are **removed** from the mixture in step 2 and in step 3?

	step 2	step 3
A	magnesium carbonate	sulfuric acid
B	magnesium carbonate	water
C	magnesium sulfate	sulfuric acid
D	magnesium sulfate	water

30 When chlorine reacts with hot NaOH(aq), the reaction below occurs.

$$3Cl_2 + 6NaOH \rightarrow 5NaCl + NaClO_3 + 3H_2O$$

How does the oxidation state of chlorine change during this reaction?

1 from 0 to –1

2 from 0 to +5

3 from 0 to + 7

A 1 only

B 1 and 2 only

C 2 and 3 only

D 1, 2 and 3

31 Which reaction does **not** involve oxidation and/or reduction?

A $2H_2 + O_2 \rightarrow 2H_2O$

B $Fe^{2+} \rightarrow Fe^{3+} + e^-$

C $2H^+ + CO_3^{2-} \rightarrow H_2O + CO_2$

D $CuO + H_2 \rightarrow Cu + H_2O$

32 Q is an oxidising agent. It is added to each of the four reagents shown in the table.

Which of the following colour change observed is correct?

	reagent	colour change
A	aqueous bromine	red brown → colourless
B	aqueous iron(III) chloride	yellow → green
C	acidified potassium manganate(VII)	purple → colourless
D	aqueous potassium iodide	colourless → brown

33 Two bottles of solution in a laboratory are unlabelled. One bottle is known to contain aqueous calcium chloride and the other, aqueous sodium sulfate.

Which reagent, when added, would help to identify the solutions?

A aqueous ammonia

B acidified barium nitrate solution

C acidified lead(II) nitrate solution

D aqueous iron(II) chloride solution

34 Which of the following statements about rubidium is **not** true?

A It has a high melting point.

B It reacts explosively with water.

C It has a higher density than potassium.

D It forms ionic compounds with non-metal elements.

35 Some properties of elements in the same group of the Periodic Table are listed.

1 charge on the ion

2 number of outer shell electrons

3 number of protons

4 total number of inner shell electrons

Which properties show an increase as the group is descended?

A 1 and 2

B 1 and 3

C 2 and 4

D 3 and 4

36 In group 17, astatine, is located below iodine.

Which statement(s) is/are correct?

1 Astatine is a more powerful oxidising agent than iodine.

2 The melting point of astatine is lower than iodine.

3 Astatine is a dark-coloured solid at room temperature.

A 1 only **B** 2 only **C** 1 and 3 **D** 3 only

37 The table shows the results of adding pieces of lead metal of known mass in salt solutions of metals P, Q and R for 15 minutes.

salt solution	initial mass of lead / g	final mass of lead after 15 minutes / g
P	5.0	3.5
Q	5.0	1.5
R	5.0	5.0

Which of the following shows the correct arrangement of metals in terms of reactivity, starting with the most reactive metal?

A Q, P, R, lead

B Q, P, lead, R

C R, lead, P, Q

D R, P, lead, Q

38 The carbonates of metals copper and Z were heated in air. Copper(II) carbonate turned from green to black when heated and carbon dioxide was given off. The carbonate of metal Z did not decompose.

Which statement(s) is/are correct?

1 Copper is more reactive than metal Z.

2 Metal Z is positioned higher in the reactivity series than copper.

3 Both metals react with hydrochloric acid to produce hydrogen gas.

A 1 only

B 2 only

C 1 and 3 only

D 2 and 3 only

39 A student made the following statements.

Statement 1 Alloys are manufactured by making a mixture of molten metals and allowing them to cool.

Statement 2 An iron alloy that contains a tiny amount of chromium is stronger than pure iron.

Which statement is true?

A Both statements are correct but statement 1 does not explain statement 2.

B Both statements are correct and statement 1 explains statement 2.

C Statement 2 is correct but statement 1 is not correct.

D Statement 1 is correct but statement 2 is not correct.

40 Some metals can be obtained by the reduction of their oxides with carbon.

Which of the following is correct?

	magnesium	silver	zinc
A	√	×	√
B	×	√	×
C	×	√	√
D	√	×	×

√ can be obtained by reduction of oxide with carbon

× cannot be obtained by reduction of oxide with carbon

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